

**gilmore-lie-groups: residuals**

equations measured 2026-09-11T10:43:57Z against the model of 2026-09-03



Flagged, not acted on

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Page | Conf.               | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Rendered                                                                                                                                                                                                                                       | Image                                                                                                                                                                                                                                                                                                                                        |
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| gilmore-lie-groups_<br>EQ0624 (1.20)                                                                                                                                                                                                                                                                                                                                                                                                                        | 24   | ■ 0.222<br>● C +21  | $\left.\left.\left(r_1-r_2\right)\right.\right\}\rightarrow\left(r_1-r_2\right),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\left(r_1-r_2\right)\overset{I}{\rightarrow}+\left(r_1-r_2\right),$                                                                                                                                                                           | $\left(r_1-r_2\right)\left\{\begin{array}{l}\overset{I}{\rightarrow}+\left(r_1-r_2\right)\\\overset{(12)}{\rightarrow}\left(r_2-r_1\right)=-\left(r_1-r_2\right)\end{array}\right.$                                                                                                                                                          |
| gilmore-lie-groups_<br>EQ0474 (8.6)                                                                                                                                                                                                                                                                                                                                                                                                                         | 145  | ■ 0.233<br>● C +114 | $\begin{gathered}\eta\left(\hat{n}+\frac{1}{2}\right)+R a^{\dagger 2}+L a^2+r a^{\dagger}+l a+\delta I \\ \mathfrak{d}\mathfrak{e}\mathfrak{f}=\left[\begin{array}{cccc} 0 & l & r & -2 \delta \\ 0 & \eta & 2 R & -r \\ 0 & -2 L & -\eta & l \\ 0 & 0 & 0 & 0 \end{array}\right] \\ \mathfrak{R}\mathfrak{e}\mathfrak{g}=\left[\begin{array}{ccc ccc} 0 & -2 R & 2 L & -r & l & 0 \\ 4 L & 2 \eta & 0 & 2 l & 0 & 0 \\ -4 R & 0 & -2 \eta & 0 & -2 r & 0 \\ \hline & & & \eta & 2 L & l \\ & & & -2 R & -\eta & -r \\ & & & 0 & 0 & 0 \end{array}\right]\begin{array}{l}\hat{n}+\frac{1}{2} \\ a^{\dagger 2} \\ a^2 \\ a^{\dagger} \\ a \\ I\end{array}\end{gathered}$ | (not rendered)                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                              |
| gilmore-lie-groups_<br>EQ0556 (10.1)                                                                                                                                                                                                                                                                                                                                                                                                                        | 174  | ■ 0.447<br>● C +25  | $\begin{aligned}&\left\{\left[H_i, H_j\right]\right\}=0 \\&\left\{\left[\mathbf{H}, E_{\alpha}\right]\right\}=\alpha E_{\alpha} \\&\left\{\left[E_{\alpha}, E_{\beta}\right]\right\}=\alpha \cdot \mathbf{H} \quad \alpha+\beta=0 \\&\quad=N_{\alpha, \beta} E_{\alpha+\beta}\end{aligned}$                                                                                                                                                                                                                                                                                                                                                                             | $\begin{aligned}&\left[H_i, H_j\right]=0 \\&\left[\mathbf{H}, E_{\alpha}\right]=\alpha E_{\alpha} \\&\left[E_{\alpha}, E_{\beta}\right]=\alpha \cdot \mathbf{H} \quad \alpha+\beta=0 \\&\quad=N_{\alpha, \beta} E_{\alpha+\beta}\end{aligned}$ | $\begin{aligned}\left[H_i, H_j\right] &= 0 \\ \left[\mathbf{H}, E_{\alpha}\right] &= \alpha E_{\alpha} \\ \left[E_{\alpha}, E_{\beta}\right] &= \alpha \cdot \mathbf{H} \quad \alpha+\beta=0 \\ &= N_{\alpha, \beta} E_{\alpha+\beta} \quad \alpha+\beta \neq 0 \text { a root} \\ &= 0 \quad \alpha+\beta \text { not a root}\end{aligned}$ |
| <b>Conf.</b> MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value<br><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ<br><b>Residual</b> render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                              |

| Identifier                                               | Page | Conf.                                                                         | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <div>gilmore-lie-groups_</div> <div>E00983 (16.26)</div> | 306  | <div><div><div></div></div>0.902</div> <div><div></div><div>C -23</div></div> | <div><math display="block">\begin{gathered} \text{monomial} \quad y^2/x \quad y/x \quad 1 \quad x \quad y \quad xy \quad x^2y \quad xy^2 \quad xy^3 \quad xy^4 \\ \begin{bmatrix} \xi_{00} &amp; \xi_{10} &amp; \xi_{01} &amp; \eta_{00} &amp; \eta_{10} &amp; \eta_{01} \\ 0 &amp; 0 &amp; -2 &amp; 0 &amp; 0 &amp; 0 \\ -1 &amp; 0 &amp; 0 &amp; 0 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; 0 &amp; +1 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; 0 &amp; 0 &amp; +2 &amp; 0 \\ 0 &amp; 0 &amp; 0 &amp; 0 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; 0 &amp; -2 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; 0 &amp; 0 &amp; -2 &amp; 0 \\ 0 &amp; -1 &amp; 0 &amp; 0 &amp; 0 &amp; -1 \\ 0 &amp; 0 &amp; -2 &amp; 0 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; +1 &amp; 0 &amp; 0 &amp; 0 \end{bmatrix} \begin{bmatrix} \xi_{00} \\ \xi_{10} \\ \xi_{01} \\ \eta_{00} \\ \eta_{10} \\ \eta_{01} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \end{gathered}</math></div> | <div><math display="block">\begin{matrix} \text{monomial} \\ y^2/x \\ y/x \\ 1 \\ x \\ y \\ xy \\ x^2y \\ xy^2 \\ xy^3 \\ xy^4 \end{matrix} \begin{bmatrix} \xi_{00} &amp; \xi_{10} &amp; \xi_{01} &amp; \eta_{00} &amp; \eta_{10} &amp; \eta_{01} \\ 0 &amp; 0 &amp; -2 &amp; 0 &amp; 0 &amp; 0 \\ -1 &amp; 0 &amp; 0 &amp; 0 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; 0 &amp; +1 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; 0 &amp; 0 &amp; +2 &amp; 0 \\ 0 &amp; 0 &amp; 0 &amp; 0 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; 0 &amp; -2 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; 0 &amp; 0 &amp; -2 &amp; 0 \\ 0 &amp; -1 &amp; 0 &amp; 0 &amp; 0 &amp; -1 \\ 0 &amp; 0 &amp; -2 &amp; 0 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; +1 &amp; 0 &amp; 0 &amp; 0 \end{bmatrix} \begin{bmatrix} \xi_{00} \\ \xi_{10} \\ \xi_{01} \\ \eta_{00} \\ \eta_{10} \\ \eta_{01} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}</math></div> | <div><math display="block">\begin{matrix} \text{monomial} \\ y^2/x \\ y/x \\ 1 \\ x \\ y \\ xy \\ x^2y \\ xy^2 \\ xy^3 \\ xy^4 \end{matrix} \begin{bmatrix} \xi_{00} &amp; \xi_{10} &amp; \xi_{01} &amp; \eta_{00} &amp; \eta_{10} &amp; \eta_{01} \\ 0 &amp; 0 &amp; -2 &amp; 0 &amp; 0 &amp; 0 \\ -1 &amp; 0 &amp; 0 &amp; 0 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; 0 &amp; +1 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; 0 &amp; 0 &amp; +2 &amp; 0 \\ 0 &amp; 0 &amp; 0 &amp; 0 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; 0 &amp; -2 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; 0 &amp; 0 &amp; -2 &amp; 0 \\ 0 &amp; -1 &amp; 0 &amp; 0 &amp; 0 &amp; -1 \\ 0 &amp; 0 &amp; -2 &amp; 0 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; +1 &amp; 0 &amp; 0 &amp; 0 \end{bmatrix} \begin{bmatrix} \xi_{00} \\ \xi_{10} \\ \xi_{01} \\ \eta_{00} \\ \eta_{10} \\ \eta_{01} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}</math></div> |
| <div>gilmore-lie-groups_</div> <div>E00490 (8.24)</div>  | 150  | <div><div><div></div></div>0.946</div> <div><div></div><div>C +20</div></div> | <div><math display="block">\frac{g}{\text{nilpotent}} = \underbrace{\left( \hat{n} + \frac{1}{2}, a^{\dagger 2}, a^2 \right)}_{\text{su}(1,1)} + \underbrace{(a^{\dagger}, a, I)}_{\text{nilpotent}}</math></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <div><math display="block">g = \underbrace{\left( \hat{n} + \frac{1}{2}, a^{\dagger 2}, a^2 \right)}_{\text{su}(1,1)} + \underbrace{(a^{\dagger}, a, I)}_{\text{nilpotent}}</math></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <div><math display="block">g = \underbrace{\left( \hat{n} + \frac{1}{2}, a^{\dagger 2}, a^2 \right)}_{\text{su}(1,1)} + \underbrace{(a^{\dagger}, a, I)}_{\text{nilpotent invariant subalgebra}}</math></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Conf.

MathPix confidence:

≥0.9

0.5-0.9

<0.5

— no value

LaTeX source

MathPix's reading, unchanged;

Rendered

the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual

render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

272 more flagged rows below the band, stated as a count: 47 component, 225 weak; confidence high 173, mid 48, low 51, none 0

Low confidence

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Page | Conf.        | LaTeX source                                                                                                                                                                                                                                                            | Rendered                                                                                                                                                                                                                                          | Image |
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| gilmore-lie-groups_<br>E00249 [conf 0.031]                                                                                                                                                                                                                                                                                                                                                                                                                  | 84   | ■ 0.031<br>● | $\begin{aligned} f_0(\theta) &= \cos \theta & \text{or } f_0(\theta) &= \cos \theta \\ f_1(\theta) &= \sin(\theta)/\theta & \theta f_1(\theta) &= \sin(\theta) \\ f_2(\theta) &= (1 - \cos(\theta))/\theta^2 & \theta^2 f_2(\theta) &= 1 - \cos(\theta), \end{aligned}$ | $f_0(\theta) = \cos \theta \quad \text{or} \quad f_0(\theta) = \cos \theta$ $f_1(\theta) = \sin(\theta)/\theta \quad \theta f_1(\theta) = \sin(\theta)$ $f_2(\theta) = (1 - \cos(\theta))/\theta^2 \quad \theta^2 f_2(\theta) = 1 - \cos(\theta)$ |       |
| <b>Conf.</b> MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value<br><b>LaTeX source</b> MathPix's reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ<br><b>Residual</b> render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured |      |              |                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                   |       |